Figure A4. Robot trajectory visualization for task "sort the objects on the table into the compost bin and the recycle bin". CoC first generates code to solve the problem, and then executes the code with Python if possible (e.g., robot APIs like `detect_objects` and `pick_place`), and with LMulator if not (e.g., commonsense QA like `is_compostable`). The robot successfully picks and places the Post-it note to the recycle bin and the orange peel to the compost bin. See the full code in Fig. A3.

```
How many countries have I been to?  I've been to Mumbai, London, Washington, Grand Canyon, Baltimore, Longsheng, Guilin, Beijing,
Galapagos, Quito, Barcelona, Paris, Prague, Nice, Dehli, Agra, Rome, Florence, Amalfi, Athens, Míkonos, Málaga, Monaco, Berlin,
Munich, Innsbruck, Bern, Milan, Lucerne, Gimmelwald (Schilthornbahn), St Moritz, St Petersburg, Helsinki, Amsterdam, Gdańsk,
Vancouver, Anchorage, Montreal, Belize, The Bahamas, Jamaica, Hawaii, Acadia National Park, Stockholm, Copenhagen, Dover, Lyon,
Madrid, Toulouse, Santorini, Oslo, Kusadasi, Souda, Rhodes, Tallinn, Venice, Naples, Cape Town, Johannesburg, Addis Abeba, Nairobi,
Seattle, San Francisco, Chicago, St Louis, Memphis, Chinle, Stanford, New York, Philadelphia, Boston, Miami, New Orleans, Walt
Disney World Resort, Jacksonville, Las Vegas, Los Angeles, Portland, Salt Lake City, Tahoe City, Phoenix, Albuquerque, Cleveland,
Charlottesville, Nags Head, Newfoundland and Labrador, Burlington, Wilmington, Myrtle Beach, St Lucia, Barbados, Banff, Haiti,
Montego Bay, Sao Palo, Rio, Lima, Cusco, Cozumel, Amarillo, Yosemite National Park, Joshua Tree, Zion National Park, Bryce Canyon
National Park, Grand Teton National Park, Yellowstone National Park, Glacier National Park, Mount Hood, Paso Robles, San Diego,
Bend, North Cascades National Park, Olympic National Park Visitor Center, Jasper National Park, Sequoia National Park, Kings Canyon
National Park, Shasta National Forest, Mount Saint Helens, Mount Rainier, Austin, Buenos Aires, El Calafate, El Chaltén, Fitz
Roy, Torres del Paine National Park, Puerto Natales, Puerto Varas, Santiago, Marble Caves, Cerro Castillo, Coyhaique, Singapore,
Casablanca, Marrakesh, Cairo, Jerusalem, Tokyo, Kyoto Prefecture, Taipei City, Taichung City, Krk, Naturpark Puez-Geisler, Ljubl-
jana, Plitvice Lakes National Park, Fairbanks, Juneau, Dallas, Sydney, Cairns, Brisbane, Hook Island, Charleston, Panama City,
Bangkok, Chiang Mai, Bengaluru, Denver, Indianapolis, Nashville, Blacksburg, Lisbon, Porto, Estes Park, Coeur d'Alene, Hood River,
Denali, Sitka, Mexico City, Warsaw, Geneva, Auckland, Queenstown, Whitefish, Minneapolis, Sioux Falls, Bozeman, Missoula, Spring-
field, Skye, Edinburgh, Honolulu, Kauai, Haleakalā National Park, Wrangell-St.  Elias National Park & Preserve, Atlanta, Tirana,
Corfu, Siena.
```

Figure A5. Full question used in Fig. 1